

ECON 7010 – Econometrics I

Lecture 1 – Course outline, intro, RStudio

Course Description

The principal objective of this course is to provide a basic introduction to econometric theory and its application. Much of the emphasis of the course is on the single-equation linear multiple regression model, under both standard and non-standard assumptions. We will be considering some of the merits and weaknesses of traditional methods of inference in the context of this model. This course is also concerned with some of the very basic approaches to identifying and handling violations of the assumptions associated with the specification of the classical multiple regression model, and some of its extensions. The course will conclude with a different estimation method: Maximum Likelihood.

Recommended Textbooks

Course notes available on the website.

- Davidson & MacKinnon, *Econometric Theory & Methods*
- Greene, *Econometric Analysis*
- [*Godwin, Introduction to Econometrics*](#) (UG level)

Course Website

Course resources (including lecture notes, past exams, assignments, and computer labs) are available on rtgodwin.com/7010

Evaluation

Assignments:	10%
Midterm Exam (Oct. 22):	30%
Final Exam:	55%

Assignments

There will be approximately 10 assignments, each worth approximately 1% of your grade. You will use RStudio and work with data in order to complete some of your assignments. All assignments must be completed independently.

Midterm and final examination

These will be closed book exams. The midterm will take place in the regular class room/time. The final exam will be scheduled by the university. The final exam is cumulative.

Grading scale

A+	92 – 100
A	85 – 92
B+	78 – 85
B	71 – 78
C+	64 – 71
C	57 – 64
D	50 – 57
F	0 – 50

- A missed assessment will result in make-up work, or reweighting of your grade.
- Nov. 24 is the last day for Voluntary Withdrawal from courses.

Academic Integrity

- All assignments and exams must be completed independently.
- Do not engage in “contract” cheating.
- Do not provide your UM Learn login information to anyone else. This is “personation”, a serious form of academic misconduct.

Ignorance is not a defense. Familiarize yourself with section 2.5 of [Academic Misconduct Procedures](#).

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ChatGPT / Claude: NO

Tentative Course Outline

- Basic Multiple Regression
- Algebraic/geometric properties of least squares (LS)
- Finite sample properties of LS
- Hypothesis Tests
- Asymptotic properties of various estimators
- Instrumental variables
- Multiple hypothesis testing
- Non-linear relationships and NLLS
- Heteroskedasticity
- ~~Autocorrelation and time series~~
- Maximum likelihood
- Bootstrapping

Student Accessibility Services

Students with disabilities should contact Student Accessibility Services to facilitate the implementation of accommodations, and meet with me to discuss the accommodations recommended by Student Accessibility Services.

Academic Supports

Sample Lecture

What is Econometrics?

- Econometrics is a subset of statistics
- Science of testing economic theories
- Used to estimate causal effects
- Used to forecast or predict (not covered in this course)
- Often characterized by “observational data”

Causal Effects

Economic models often suggest that one variable causes another. This often has *policy implications*. The economic models, however, do not provide *quantitative magnitudes* of the causal effects.

For example:

- how does another year of *education* change *earnings*?
- How does a change in *interests rates* affect *inflation*?

Using data to estimate causal effects

An experiment would be best.

- How would you determine the effect of fertilizer on crop yield?
- How would you use an experiment to determine the above causal effects?
- What is the advantage of experiments?

Economic experiments are usually unethical and/or too expensive.

We usually don't have *experimental* data in econometrics – we have *observational* data.

There are issues when dealing with observational data:

- Omitted variables
- Simultaneous causality
- Correlation vs. causation

These issues lead to the need for what applied econometricians refer to as an *identification strategy*.

Objectives of this course

- Learn about OLS
- Learn about the OLS assumptions
- Re-evaluate the assumptions – learn new methods
- Learn about hypothesis testing

R and RStudio

The theory and concepts presented in this course will be illustrated by analysing several data sets. Data analysis will be accomplished through the R Statistical Environment and RStudio. Both are free, and R is fast becoming the best and most widely used statistical software.

First, install R

- Go to <https://muug.ca/mirror/cran/>
- Choose Windows or Mac

← → ↻ muug.ca/mirror/cran/ ☆ ⚙ R ⋮



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The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux](#)
- [Download R for \(Mac\) OS X](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2020-06-22, Taking Off Again) [R-4.0.2.tar.gz](#), read [what's new](#) in the latest version.
- Sources of [R alpha and beta releases](#) (daily snapshots, created only in

- Click “install R for the first time”

← → ↻ muug.ca/mirror/cran/ ☆ ⚙ R ⋮



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R for Windows

Subdirectories:

base	Binaries for base distribution. This is what you want to install R for the first time .
contrib	Binaries of contributed CRAN packages (for R >= 2.13.x; managed by Uwe Ligges). There is also information on third party software available for CRAN Windows services and corresponding environment and make variables.
old contrib	Binaries of contributed CRAN packages for outdated versions of R (for R < 2.13.x; managed by Uwe Ligges).
Rtools	Tools to build R and R packages. This is what you want to build your own packages on Windows, or to build R itself.

Please do not submit binaries to CRAN. Package developers might want to contact Uwe Ligges directly in case of questions / suggestions related to Windows binaries.

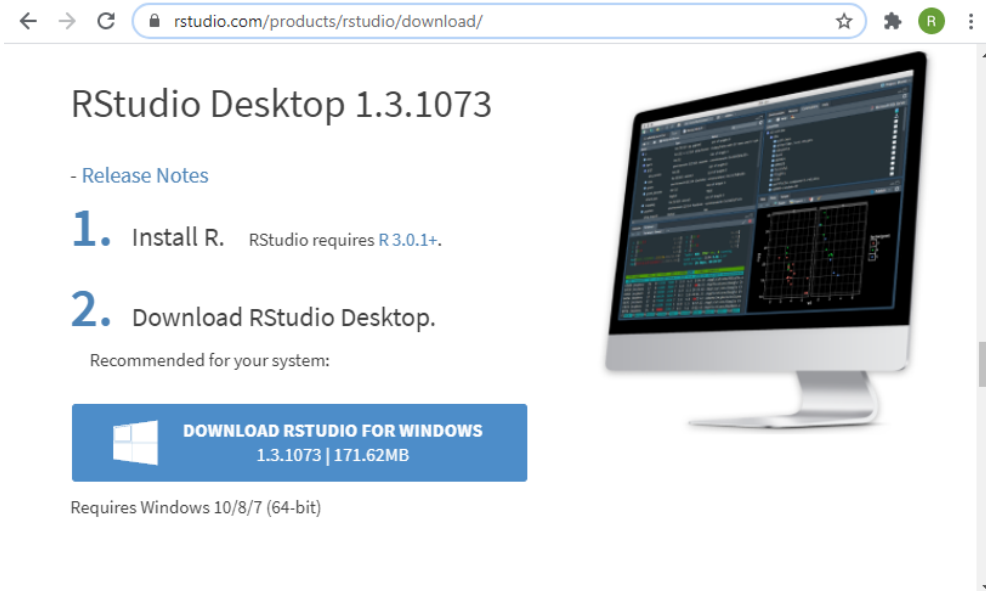
You may also want to read the [R FAQ](#) and [R for Windows FAQ](#).

Note: CRAN does some checks on these binaries for viruses, but cannot give guarantees. Use the normal precautions with downloaded executables.

- Click “Download R 4.0.2 for Windows” (or Mac)
- Run the “.exe” file
- Click “Next” a bunch of times

Second, install RStudio

- Go to <https://rstudio.com/products/rstudio/download/>
- Scroll down until you see the download button “Download RStudio for Windows (Mac)”. Click it.



- Run the “.exe”
- Keep clicking “next” / “install”
- Find RStudio on your computer and open it. It should look something like this:

```
~/  
R version 4.0.2 (2020-06-22) -- "Taking off Again"  
Copyright (C) 2020 The R Foundation for Statistical Computing  
Platform: x86_64-w64-mingw32/x64 (64-bit)  
  
R is free software and comes with ABSOLUTELY NO WARRANTY.  
You are welcome to redistribute it under certain conditions.  
Type 'license()' or 'licence()' for distribution details.  
  
Natural language support but running in an English locale  
  
R is a collaborative project with many contributors.  
Type 'contributors()' for more information and  
'citation()' on how to cite R or R packages in publications.  
  
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.  
  
> |
```

Environment is empty

	Name	Size	Modified
☐	Custom Office Templates		
☐	desktop.ini	402 B	Sep 7, 2020, 12:47 PM
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